

Part A Chapter 5

A Society Based on Decentralised Decision Making: Selforganisation in Honeybee Societies¹

1 Introduction

When an individual functions in a dynamic environment, he or she often needs to have the ability to adapt to environmental changes, to ensure a continuation of proper functioning. The same holds for groups of individuals. However, in the case of a group, proper coordination of the individual actions becomes an issue. Such adaptations can be made in two different ways. One way is through a centralised decision process, for example, in the form of a hierarchical organisation structure in which some leader or manager decides and tells the others what to do. Another way is through a decentralised process in which the adaptation is based on decisions by each the individuals. Out of the combination of these individual decisions a form of (self)organisation emerges; see also (Camazine, Deneubourg, Franks, Sneyd, Theraulaz, and Bonabeau, 2001; Bonebeau and Theraulaz, 2000). For social insects, such as honeybees and wasps, their societies are known to adapt in a decentralized fashion to environmental changes. Thus, on the basis of individual decision processes they create and maintain a wellfunctioning society.

In human society similar patterns can be observed, often to the benefit of society as a whole. Examples are taking initiatives to create businesses, or designing new products. In this chapter a honeybee colony is analysed as an example of a form of selforganisation based on individual decision making. The model presented is a variation on the model described in (Hoogendoorn, Schut, and Treur, 2007).



¹ Based on: Hoogendoorn, M., Schut, M.C., and Treur, J., Modeling Decentralized Organizational Change in Honeybee Societies. In: Costa, F.A., Rocha, L.M., Costa, E., Harvey, I., and Coutinho, A. (eds.), *Advances in Artificial Life, Proc. of the 9th European Conference on Artificial Life, ECAL'07*. Lecture Notes in Computer Science, vol. 4648. Springer Verlag, 2007, pp. 615-624.

1.1 Domain description and focus

Within societies of honeybees a number of different roles for bees can be distinguished, as is shown in the literature; see e.g., (Schultz, Barron, and Robinson, 2002; Winston and Punnet, 1982). For the sake of brevity only four will be addressed here:



- a *brood carer* takes care of feeding the larvae within the bee hive
- a *patroller* guards the hive by killing enemies entering the hive
- a *forager* harvests food to be stored in the hive
- an *undertaker* cleans the hive of corpses

In principle every bee can perform any of these roles. Switching between roles is triggered by changes

in the environment observed by the bees. Such observations differ per bee. Each role has a specific trigger, for which a bee has a certain threshold that determines whether this is the role it should play. The bee always plays the role for which it is most triggered, relative to its threshold. For example, bees are triggered to start playing the (*brood*) *carer* role when they observe the larvae emitting a too high level of hunger pheromones, a chemical substance indicating their need for food. Once they are allocated to the role, they start getting food from the combs and feed the larvae that are emitting the pheromones. A trigger for the *patroller* role is the amount of enemies observed around the hive. *Foragers* that have returned from their hunt for food, communicate the location where they found the food by means of the honeybee dance. Other bees currently not playing the *forager* role, can be triggered to start playing the *forager* role. The more corpses there are, the more bees are being triggered to switch from their current role to being *undertaker*.



For the bee society important aspects considered are *larvae strength*, *safety*, *food availability*, and *cleanness* (addressed, respectively, by the brood carer, patroller, forager, and undertaker roles). For example, for wellfed brood, this concerns relative larvae hunger, indicated by the larvae pheromone rate, which should be as low as possible. In order to maintain these aspects, a certain effort performed by the bees is needed all the time. This effort has to address the most urgent aspects in the first place, but should do so without completely neglecting other

aspects with urgency a bit lower. For instance, the bees in the brood carer role take care that the larvae are fed. The effort to maintain the hunger of larvae at a low level is performed by feeding the larvae.

Whether the society will function well, depends on the flexibility of the role allocations. For example, if the number of larvae or enemies increases, also the number of brood carers, respectively patrollers should increase. If the adaptation to the new situation takes too much time, this will damage the society. Therefore, an adaptation mechanism is needed for the society that is sufficiently flexible. This adaptation mechanism of the society as a whole works by selforganisation based on decisions made by individuals. The individual decision making is assumed to be based on norms that are used as a kind of thresholds; e.g., see (Theraulaz, Bonabeau, and Deneubourg, 1998):

- Each individual maintains norms for each of the four aspects to be maintained.
- For each aspect, the individual determines in how far the current state is unsatisfactory, expressed in a degree of urgency for that aspect.
- For each aspect the relative urgency is determined (relative to the individual's norm for that aspect): the degree of urgency divided by the norm for that aspect
- The most urgent aspect is the one with highest relative urgency.
- The role is taken that relates to the most urgent aspect.
- Norms are adapted: if an individual is in a role addressing a certain aspect, then the norm for that aspect is slowly decreased over time, whereas the norms for the other aspects are increased accordingly, keeping the total sum of the norms at 1.

The last item makes that an individual may become more inclined to choose certain aspects: those with which it has built up has more experience. This mechanism may lead to the emergence of some forms of specialisation within society. This pattern can also be observed in human society: not seldom people choose for tasks that are a bit familiar.

1.2 Characteristic patterns to be explored

Given the above considerations, the following patterns can be explored by a model:

- In how far are the society's parameters kept between certain bounds by the individual decision making processes (*homeostasis*); in particular, to which extent are wellfed brood, safety, food storage, and cleanness maintained.
- How long does it take to address sudden threats of the society's parameters; in how far are fast changing environments a problem.
- In how far keep individuals their roles over longer periods; in how far is specialisation of individuals emerging in the process.

2 Analysis of the Main Aspects and Their Relations

This section describes globally the relevant concepts used and their relations.

2.1 Identification of relevant concepts

To describe the bee society a number of concepts are relevant. First of all are the *bee individuals* and the *roles* (*carer*, *forager*, *patroller*, and *undertaker*) crucial, and in addition, the *allocations* of roles to individuals. For a given allocation, a certain *effort* is provided.

Each of the roles relates to an *aspect* of the society that has to be maintained (respectively *food availability*, *larvae strength*, *cleanness*, and *safety*). At each point in time, each of these aspects of the society has some *level*, which should be kept low; the *urgency* of an aspect is indicated by its level (high level is high urgency). Notice that this (time-dependent) aspect level or urgency is an objective concept, for the society as a whole, not dependent on which individual is considering it. The specific aspects are related to the amounts of *larvae*, *stored food*, *enemies*, and *deaths*.

For each individual and each aspect to be maintained there is a *norm* that is used as a threshold for the urgency of that aspect, to get a *relative urgency* for the aspect. Both the norms and the relative urgencies are (time-dependent) subjective concepts, depending on the individual. Norms are adapted over time depending on the individual's allocations, according to a *norm adaptation factor*. In summary, the following concepts are used to describe the bee society:

- bee individuals
- different roles: forager, carer, undertaker, and patroller role
- different aspects of the society to be maintained, and the levels of them: food availability, larvae strength, cleanness, and safety
- for each individual and aspect: a norm that is to be used as threshold to determine the relative urgency
- maximal relative urgency for each individual
- norm adaptation factor
- role allocations to individuals
- effort made, work contributed
- observed world
- amounts of larvae, enemies, deaths

2.2 Overall structure of the model

The model is structured according to three levels and their interlevel relations:

- the *collective societal level* of the society as a whole,
- the (*external*) *individual behavioural level* of the various individual behaviours, and
- the decision making processes at the (*internal*) *cognitive level*.

At the collective societal level the functioning of the society is described: the development of society over time, for example, with respect to food availability, larvae strength, safety, cleanness. At the individual behavioural level the individual behaviours from an external viewpoint are described: the work delivered. At the internal cognitive level, the decision making processes are described: based on norms and estimated urgencies. Interaction between the three levels takes place in two directions:

- *Upward interaction from individual behaviour to societal*
The individual behaviours (contributed work) affect the society's processes and states concerning food availability, safety, cleanness and larvae strength.
- *Downward interaction from societal process to individual behaviour*
The society's states and processes over time with respect to the relevant aspects (cleanness, safety, food availability, and larvae strength) are observed as input for the individual behaviours.
- *Upward interaction from internal cognitive process to individual behaviour*
The internal decision making process for role allocation generates the individual behaviour: the work contributed.
- *Downward interaction from individual behaviour to cognitive process*
Observations (of cleanness, safety, food availability, and larvae strength) at the individual behavioural level affect the internal cognitive decision making processes for role allocation.

This model structure shows an example of how phenomena at the society or collective level can interact with processes of individuals. Note that in this specific case it is assumed that no direct interaction at the local level takes place (e.g., no communication). However, for other cases that can be added to the model.

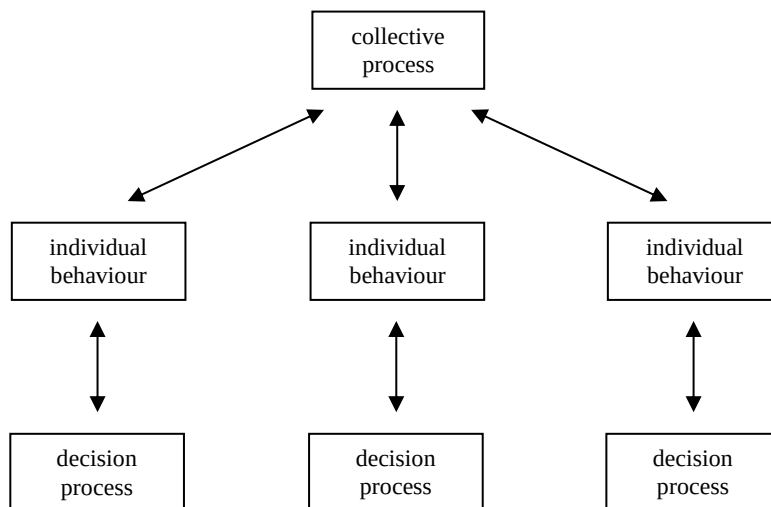


Figure 1 Overall structure of the model: cognitive, behavioural and collective level

2.3 Relationships between the identified concepts

The temporal relationships between the identified concepts are as follows.

- levels of food need, larvae weakness, uncleanness, and unsafety affect world observations
- world observations and individual norms for carer, forager, patroller, and undertaker role affect allocations of individuals to roles
- the norm adaptation factor and the allocations of an individual to a role affect its norms for carer, forager, patroller, and undertaker roles
- allocations of individuals to roles and the effort made affect the contributed work
- the contributed work affects the levels of food need, larvae weakness, uncleanness, and unsafety
- the number of enemies affects the level of unsafety
- the number of deaths affects the level of uncleanness
- the number of larvae affects larvae weakness and food need

In graphical form these relationships are depicted in Figure 2.

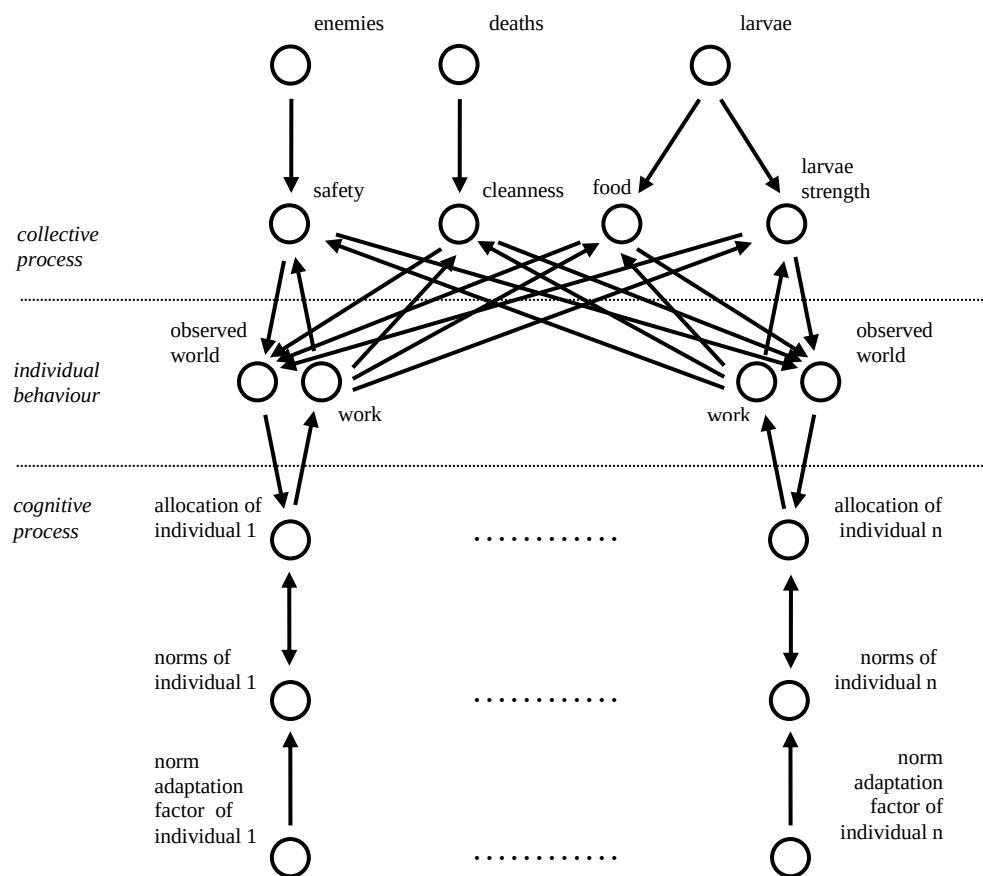


Figure 2 Dynamic relationships in the model

3 The Detailed Model

In this section the detailed model is described, by formalising the relevant concepts and their dynamic relations in a detailed manner.

3.1 Formalising the concepts used

To obtain a detailed model, first the concepts used are formalised. In this case for each of the concepts a numerical variable is introduced that takes natural or real numbers as values; see Table 1. For the sake of simplicity the observations of levels of food availability, cleanness, et cetera, are not distinguished from the levels themselves.

forager, carer, undertaker, and patroller role	<i>1, 2, 3, 4</i>
individuals	<i>Bee1, Bee2,</i>
role allocations between individuals and roles	<i>Bee1Role, Bee2Role, ...</i>
for each individual: norms used as thresholds for carer, forager, patroller, and undertaker role	<i>Bee1ForagerNorm, Bee1CarerNorm, Bee1UndertakerNorm, Bee1PatrollerNorm, ...</i>
maximal urgency for each individual	<i>Bee1MaxUrgency, Bee2MaxUrgency,</i>
norm adaptation factor for individuals	<i>Bee1AdapFactor, Bee2AdapFactor, ...</i>
effort per individual	<i>Effort</i>
food need, larvae weakness, uncleanness, unsafety	<i>FoodNeed, LarvaeWeakness, Uncleanness, Unsafety</i>
contributed work	<i>ForagingWork, CaringWork, PatrollingWork, CleaningWork</i>
number of (new) enemies	<i>NewEnemies</i>
number of (new) deaths	<i>NewDeaths</i>
the number of larvae	<i>Larvae</i>

Table 1 Formalisation of the relevant concepts

3.2 Formalising the relationships

The dynamics of these concepts involve temporal relationships, which are analysed in more detail below. They describe the basic parts of the process; see also Figure 2. Notice that, compared to Figure 2, the process to determine allocation of roles is split here into two steps.

From levels of food need, larvae weakness, cleanness, and safety, and individual norms to maximal relative urgency

First relative urgency of each of the four aspects is determined by dividing the level by the norm for that aspect. Next the maximal relative urgency is determined. This is done for each individual. For example, for Bee1:

$$Bee1MaxUrgency(t) = \max(FoodNeed(t)/Bee1ForagerNorm(t), LarvaeWeakness(t)/Bee1CarerNorm(t), Uncleanness(t)/Bee1UndertakerNorm(t), Unsafety(t)/Bee1PatrollerNorm(t))$$

From maximal relative urgency to allocations of individuals to roles

The role at the next time point is the one for the aspect with highest relative urgency, for any of the individuals.

$$\begin{array}{llll}
 Bee1Role(t + \Delta t) & = & 1 & \text{if } FoodNeed(t)/Bee1ForagerNorm(t) = Bee1MaxUrgency(t) \\
 & & 2 & \text{if } LarvaeWeakness(t)/Bee1CarerNorm(t) = Bee1MaxUrgency(t) \\
 & & 3 & \text{if } Uncleanness(t)/Bee1UndertakerNorm(t) = Bee1MaxUrgency(t) \\
 & & 4 & \text{if } Unsafety(t)/Bee1PatrollerNorm(t) = Bee1MaxUrgency(t)
 \end{array}$$

From adaptation factor and allocation of an individual to a role to its norms

Norms for a given individual are adapted by multiplying the norm for the current role by the adaptation factor. To get the sums of all norms at 1 again, they are divided by an appropriate number.

If $BeelRole(t) = 1$

$$\begin{aligned} BeelForageNorm(t + \Delta t) &= c * BeelAdapFactor * BeelForagerNorm(t) \\ BeelCareNorm(t + \Delta t) &= c * BeelCarerNorm(t) \\ BeelUndertakeNorm(t + \Delta t) &= c * BeelUndertakerNorm(t) \\ BeelPatrolNorm(t + \Delta t) &= c * BeelPatrollerNorm(t) \end{aligned}$$

where c is a factor that takes care that the sum of the new norms becomes 1 again:

$$c = 1 / (1 - (1 - BeelAdapFactor) * BeelForageNorm(t))$$

If $BeelRole(t) = 2$

$$\begin{aligned} BeelCareNorm(t + \Delta t) &= c * BeelAdapFactor * BeelCarerNorm(t) \\ BeelForageNorm(t + \Delta t) &= c * BeelForagerNorm(t) \\ BeelUndertakeNorm(t + \Delta t) &= c * BeelUndertakerNorm(t) \\ BeelPatrolNorm(t + \Delta t) &= c * BeelPatrollerNorm(t) \end{aligned}$$

where c is a factor that takes care that the sum of the new norms becomes 1 again:

$$c = 1 / (1 - (1 - BeelAdapFactor) * BeelCareNorm(t))$$

If $BeelRole(t) = 3$

$$\begin{aligned} BeelUndertakeNorm(t + \Delta t) &= c * BeelAdapFactor * BeelUndertakerNorm(t) \\ BeelForageNorm(t + \Delta t) &= c * BeelForagerNorm(t) \\ BeelCareNorm(t + \Delta t) &= c * BeelCarerNorm(t) \\ BeelPatrolNorm(t + \Delta t) &= c * BeelPatrollerNorm(t) \end{aligned}$$

where c is a factor that takes care that the sum of the new norms becomes 1 again:

$$c = 1 / (1 - (1 - BeelAdapFactor) * BeelUndertakeNorm(t))$$

If $BeelRole(t) = 4$

$$\begin{aligned} BeelPatrolNorm(t + \Delta t) &= c * BeelAdapFactor * BeelPatrollerNorm(t) \\ BeelForageNorm(t + \Delta t) &= c * BeelForagerNorm(t) \\ BeelCareNorm(t + \Delta t) &= c * BeelCarerNorm(t) \\ BeelUndertakeNorm(t + \Delta t) &= c * BeelUndertakerNorm(t) \end{aligned}$$

where c is a factor that takes care that the sum of the new norms becomes 1 again:

$$c = 1 / (1 - (1 - BeelAdapFactor) * BeelPatrolNorm(t))$$

Determining the level of food need

The level each of the aspects is decreased by the amount of work spent on that aspect, and increased by a number that depends on the circumstances (numbers of larvae, enemies, deaths, individuals). These circumstances are factors external to the model. They are set by hand in the form of scenarios. If the outcome would be below 0, the number 0 is taken for the level. In particular, for the level of food need the addition is just the sum of the numbers of larvae and bees.

$$FoodNeed(t + \Delta t) = FoodNeed(t) + FoodneedAddition(t) - ForagingWork(t)$$

with

$$ForagingWork(t) = Effort * number\ of\ individuals\ with\ role\ allocation\ Forager$$

$$FoodneedAddition(t) = Larvae(t) + number\ of\ individuals$$

Determining the level of larvae weakness

Also the level of this aspect is decreased by the amount of work spent on it, and increased by a number that depends on the circumstances. For the level of larvae weakness, the addition is the number of larvae, as these are to be fed in the next period. If the outcome would be below 0, the number 0 is taken for the level.

$$\text{LarvaeWeakness}(t+\Delta t) = \text{LarvaeWeakness}(t) + \text{LarvaeWeaknessAddition}(t) - \text{CareWork}(t)$$

with

$$\text{CareWork}(t) = \text{Effort} * \text{number of individuals with role allocation Carer}$$

$$\text{LarvaeWeaknessAddition}(t) = \text{Larvae}(t)$$

Determining level of uncleanness

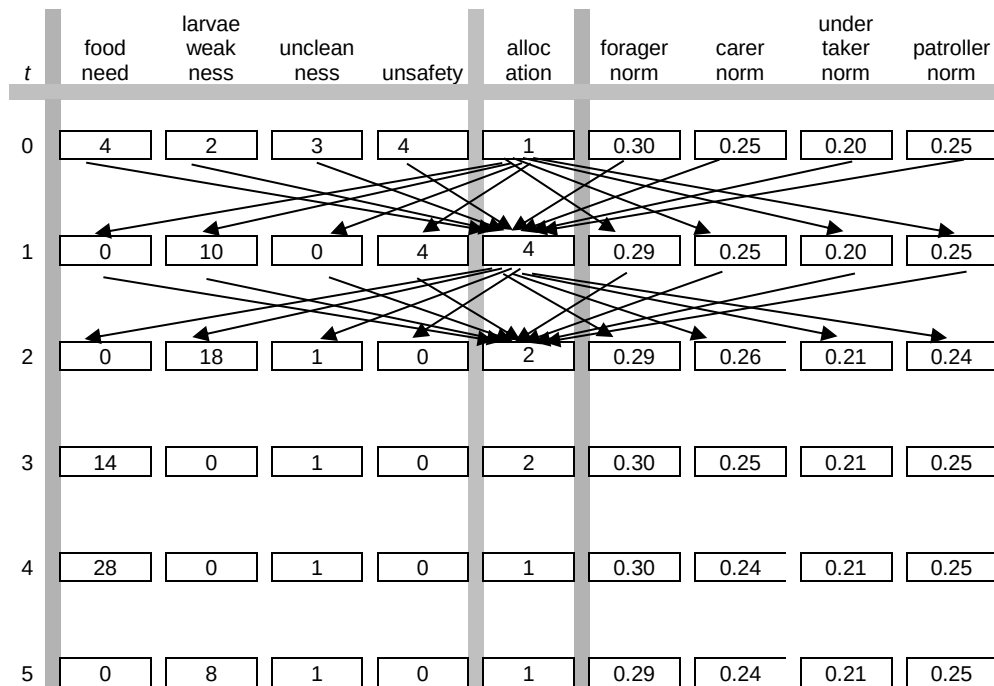
As for the aspects above, the level of this aspect is decreased by the amount of work spent on that aspect, and increased by a number that depends on the circumstances. In this case the number of bees that just died adds to the level of uncleanness. If the outcome would be below 0, the number 0 is taken for the level.

$$\text{Uncleanness}(t+\Delta t) = \text{Uncleanness}(t) + \text{UncleannessAddition}(t) - \text{UndertakerWork}(t)$$

with

$$\text{UndertakerWork}(t) = \text{Effort} * \text{number of individuals with role allocation Undertaker}$$

$$\text{UncleannessAddition}(t) = \text{NewDeaths}(t)$$

**Table 2 Dependencies between the different concepts over time****Determining of the level of unsafety**

Finally, also for the aspect of unsafety, the level is decreased by the amount of work spent on that aspect, and increased by a number that depends on the circumstances. Here the number of newly arriving enemies is taken as the addition to unsafety. If the outcome would be below 0, the number 0 is taken for the level.

$$\text{Unsafety}(t+\Delta t) = \text{Unsafety}(t) + \text{UnsafetyAddition}(t) - \text{PatrolWork}(t)$$

with

$$\text{PatrolWork}(t) = \text{Effort} * \text{number of individuals with role allocation Patrol}$$

$$\text{UnsafetyAddition}(t) = \text{NewEnemies}(t)$$

In Table 2 based on these relationships the dependencies between the values are shown for an example scenario. The four columns at the right hand side show the norms of one of the bees, the middle column shows the allocation (for that bee), and the four columns at the left hand side show the indicator aspects of the society. It is shown that all values of these aspects and of the norms affect the allocation and vice versa.

4 Simulation Experiments

A number of simulation experiments have been performed in Microsoft Excel. In Figures 3 to 5 results are shown of one of these experiments. To keep it a bit simple, only two bees are considered.

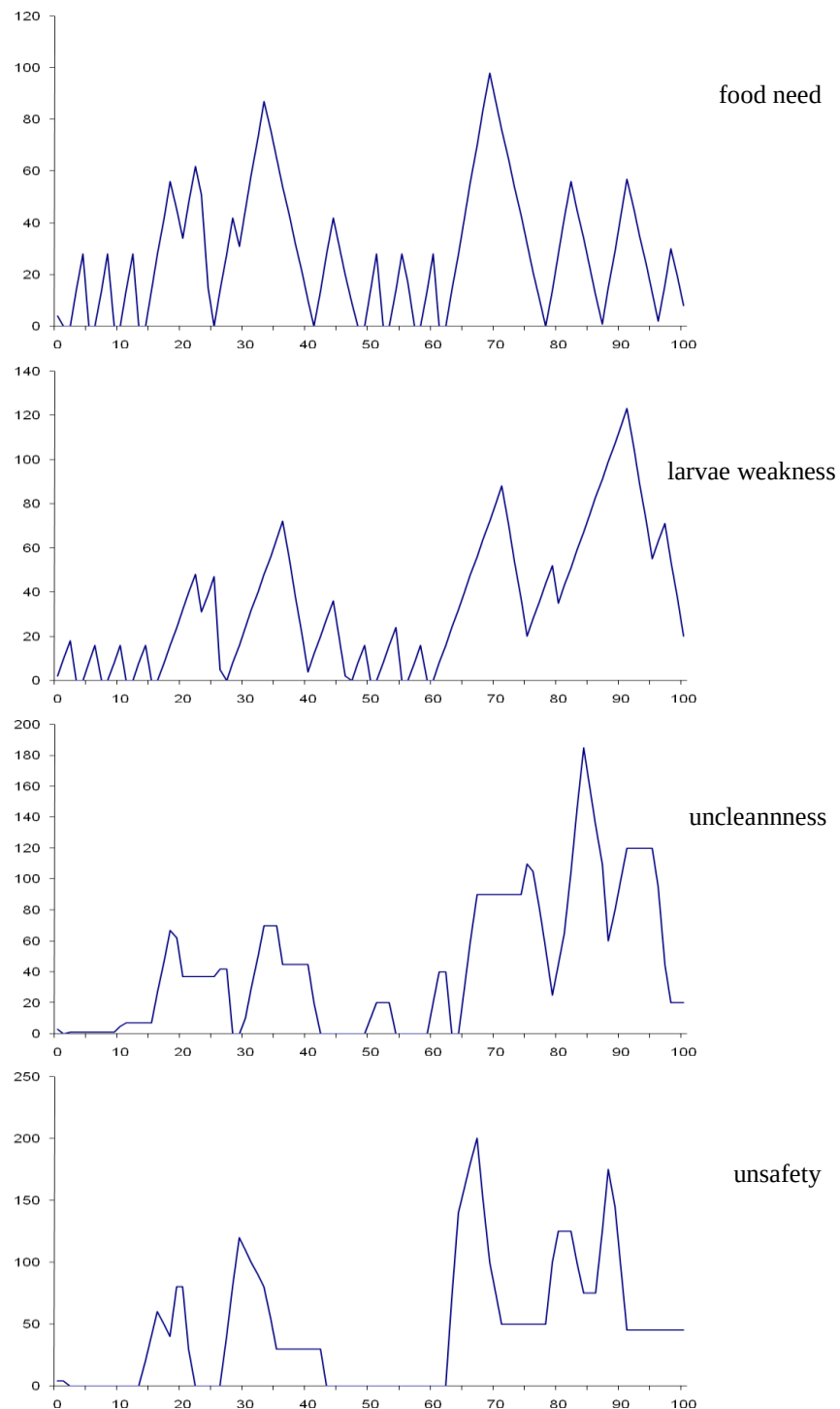


Figure 3 Example simulation trace: indicators for the society's wellbeing

First, in Figure 3 the society's wellbeing indicators are shown. Here a low value means a good (non-threatening) situation. The values should not increase beyond all bounds: in that case the situation would not be under control anymore. As can be seen in Figure 3 this is achieved: all the time a value goes up, it is decreased to a low level by working on it. The role allocations of the two bees are shown in Figure 4. What can be seen here, for example, is that bee 2 often takes the role of forager, in contrast to bee 1; bee 1 is more often involved in feeding the larvae, where bee 2 after some initial period refrains from taking that role. In fact, it seems that bee 2 has specialised being most of the time a patroller and forager, whereas bee 1 does all roles from time to time.

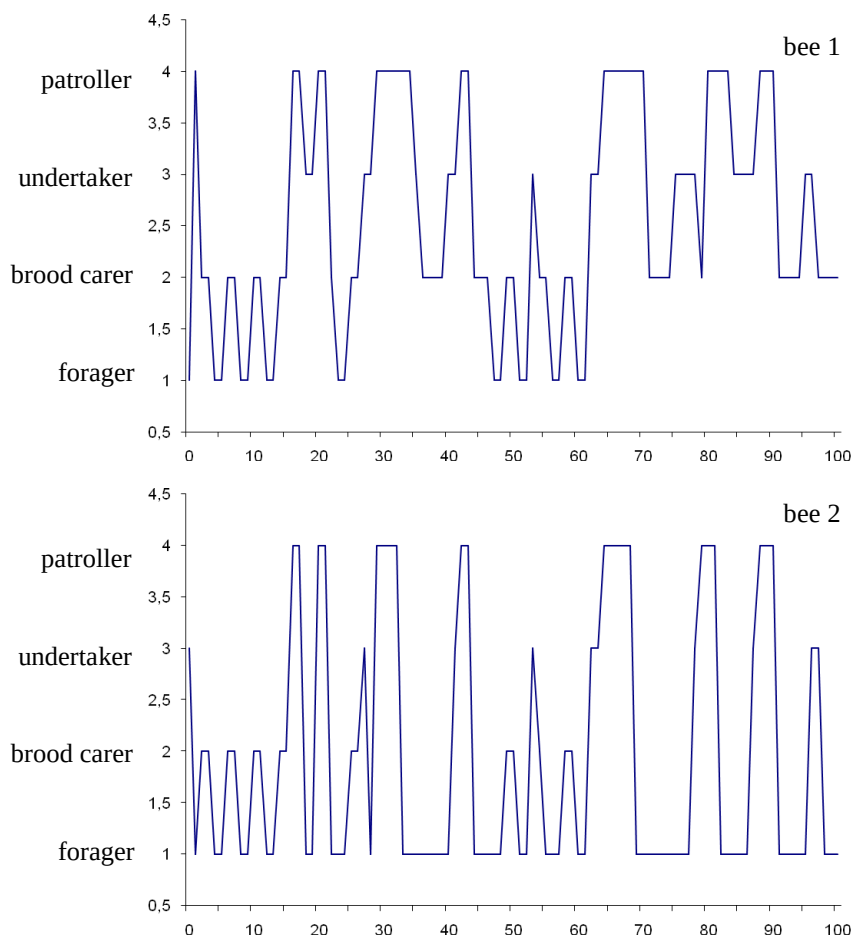


Figure 4 Example simulation trace: roles over time

Figure 5 shows how for the two bees the norms were adapted over time. Also here a clear trend in specialisation is visible, in particular for foraging, for which bee 1 developed a high

threshold, whereas bee 2 developed a very low threshold. Moreover, for brood care bee 1 develops a low threshold and bee 2 a rather high threshold.

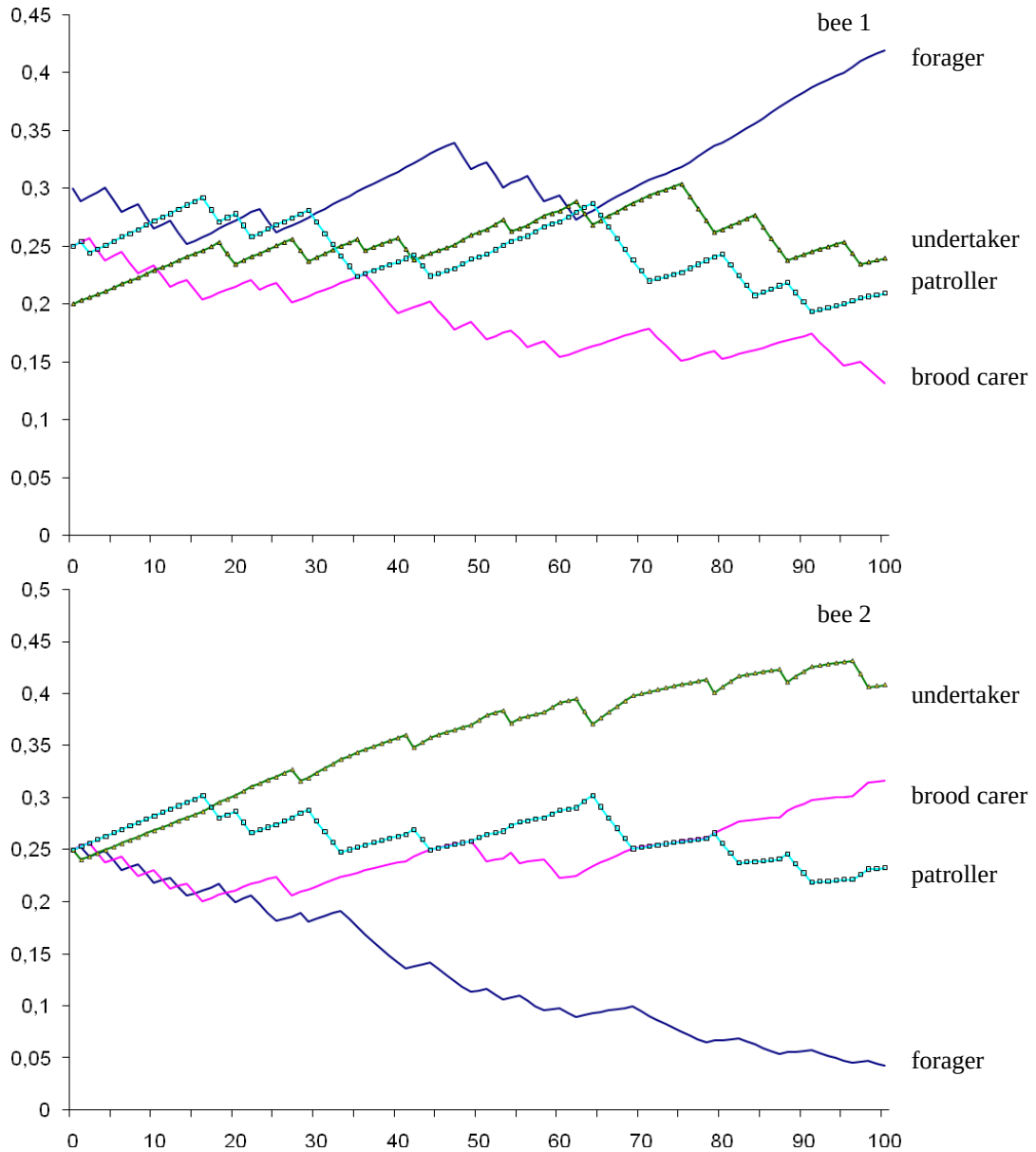


Figure 5 Example simulation trace: specialisation by adaptation of norms

5 Evaluation

In this section, the simulations performed using the model are evaluated with respect to the characteristic patterns identified in Section 1.2. Moreover, the assumptions behind the model are summarised. Finally, possibilities of modification or refinement of the model are discussed.

5.1 Evaluation of the simulations and characteristic patterns

The following characteristic patterns were formulated in Section 1.2, and were explored by the model.

In how far are the society's parameters kept between certain bounds (homeostasis)

The experiments show that indeed these parameters are kept within certain bounds. However, this is depending on assumptions made on the environment, namely that the total amount of work needed per time unit does not exceed the total effort per time unit which can be provided by the total population of bees. For example, if the total effort per time unit which can be provided by the total population of bees is 200, and every time unit 400 enemies show up, then the society will certainly not stay within bounds: the levels will raise beyond any bound and the situation will be out of control.

How long does it take to address sudden threats of the society's parameters; in how far are fast changing environments a problem.

For the environment scenarios used the changes were taken care of in reasonably short times. Scenarios with very fast changes still can be tried.

In how far is specialisation of individuals emerging in the process.

Indeed in most of the simulations some form of specialisation emerges during the process. Although the general idea is that in principle all bees can do all roles, in practice they often develop strong tendencies to take certain roles and to avoid other roles.

5.2 Assumptions behind the model

The assumptions made include:

- Every bee can play every role.
- All individuals have the same global information about the society
- Individuals may have different characteristics, and therefore differ in their role allocation behaviour
- Decision making takes place individually.
- Bees have a slight preference to stay in a role.
- Number of larvae, enemies and deaths are independent, external factors, not dependent on, for example, number of bees or amount of food available, or dependent on each other.

5.3 Possible variations and refinements

Variations and refinements of the model, for example, can be made in the following manners.

- The number of bees can be increased.
- The norm adaptation factor can be varied more, and possibly made different for different bees.
- The environmental dynamics can be varied more.
- Individuals can be given different types of information about the society, for example, depending on their position, or depending on their subjective estimation.
- Birth of new bees and death of old bees can be handled more explicitly, as well as the transition from larvae to bees. For example, the number of deaths depends on number of enemies around.

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